UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2012 question paper

for the guidance of teachers

9709 MATHEMATICS

9709/12 Paper 1, maximum raw mark 75

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- Cambridge will not enter into discussions or correspondence in connection with these mark schemes.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

- The symbol √ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.

- Note: B2 or A2 means that the candidate can earn 2 or 0.
  B2/1/0 means that the candidate can earn anything from 0 to 2.

  The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.

- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking *g* equal to 9.8 or 9.81 instead of 10.

The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only - often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

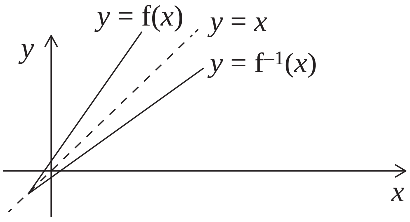
Penalties

MR -1	A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through √" marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR-2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA -1	This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

1 $y = \dfrac{6}{2x-3}$		
Integral of $y^2 = \dfrac{-36}{(2x-3)} \div 2$	B1 B1	co allow 2nd B1 independent of 1st.
Use of limits 2, 3	M1	Used as 2 to 3 or 3 to 2 in integral of y^2.
$\rightarrow$ 12π or 37.7	A1	Co
	[4]	(uses area 0/4. No π Max 3/4)
2 $y = 4\sqrt{x} + \dfrac{2}{\sqrt{x}}$		
(i) $\dfrac{dy}{dx} = 4.\tfrac{1}{2}.x^{-\frac{1}{2}} + 2.(-\tfrac{1}{2})x^{-1.5}$	M1	Reducing "their" power by 1 once. Allow unsimplified
or $y = \dfrac{2}{\sqrt{x}} - \dfrac{1}{x^{1.5}}$	A1 A1	
	[3]	
(ii) $\dfrac{dy}{dt} = \dfrac{dy}{dx} \times \dfrac{dx}{dt}$ used	M1	Must be used correctly
$\rightarrow$ $= \tfrac{7}{8} \times 0.12 = 0.105$	A1	co – fraction or decimal.
	[2]	
3 Coeff of x^3 in $(a+x)^5 = 10 \times a^2$	B1 B1	co co
Coeff of x^3 in $(2-x)^6 = -160$	B1	co
$\rightarrow$ $10a^2 - 160 = 90$	M1	forms an equation from 2 terms
$\rightarrow$ $a = 5$	A1	co
	[5]	
4 $A(-1, -5)$, $B(7, 1)$.		co
M (3, −2)	B1	
Gradient = $\frac{3}{4}$		co
Perpendicular gradient = $-\frac{4}{3}$	B1	needs to be perp. through *M*.
Eqn $y + 2 = -\frac{4}{3}(x-3)$	M1	Setting **one** of x or y to 0.
Sets x and y to 0 $C(\frac{3}{2}, 0)$ $D(0, 2)$	M1	Correct method. co.
$\rightarrow$ Pythagoras $\rightarrow$ CD = 2.5	M1A1	
	[6]	

5 $\tan x + \frac{1}{\tan x} \equiv \frac{1}{\sin x \cos x}$		
(i) LHS = $\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}$	M1	Use of tan = sin/cos twice
= $\frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x}$	M1 [2]	Use of $s^2 + c^2 = 1$ appropriately – everything correct.
(ii) $\frac{2}{\sin x \cos x} = 3\tan x + 1$		
Uses (i) $2(\tan x + \frac{1}{\tan x}) = 3\tan x + 1$	M1	Uses part (i) to obtain eqn in tan$x$ only
$\rightarrow \tan^2 x + \tan x - 2 = 0$	DM1	Correct soln of quadratic eqn
$\rightarrow$ tan$x$ = 1 or −2	B1 A1	co. Must have correct quadratic co
$\rightarrow x = 45°$ or $116.6°$	[4]	
6 **(i)** cosine rule or $2 \times r \times \sin \frac{1}{2}.(2.4)$	M1	Any complete valid method.
$\rightarrow$ 14.9 cm	A1 [2]	co
(ii) Perimeter = (i) $+ r\theta$	M1	Uses $s = r\theta$ with 2.4, or $\pi - 2.4$,
$\theta = 2\pi - 2.4$,	B1	or $2\pi - 2.4$
$\rightarrow$ 46.0 cm	A1√	Anywhere in parts (ii) or (iii).
	[3]	Adds 31.1 to (i) for √.
(iii) Area = Sector + triangle		
$\frac{1}{2} \times 8^2 (2\pi - 2.4) + \frac{1}{2} \times 8^2 \sin 2.4$	M1 M1	
124.3 + 21.6 $\rightarrow$ 146 cm².	A1	Uses $\frac{1}{2}r^2\theta$. Uses any valid method.
	[3]	co
7 **(a)** $S_n = n^2 + 8n$.		
$S_1 = 9 \rightarrow a = 9$	B1	co
$S_2 = 20 \rightarrow a + d = 11 \rightarrow d = 2$	M1 A1	Realises that S_2 is $a + (a + d)$. co
(or equating $n^2 + 8n$ with S_n and comparing coefficients)	[3]	
(b) $a - ar = 9$	B1	co
$ar + ar^2 = 30$	B1	co
Eliminates a $\rightarrow 3r^2 + 13r - 10 = 0$	M1	Complete elimination of r or a
or $\rightarrow 2a^2 - 57a + 81 = 0$		Correct quadratic.
$\rightarrow r = \frac{2}{3}$	A1	
$\rightarrow a = 27$	A1 [5]	co (condone 27 or 1.5)

8 **(i)** 3**i** − 4**k**, 2**i** + 3**j** − 6**k**. Dot product = 6 + 24 = 30 = $\sqrt{25} \times \sqrt{49}$ cos θ → angle = 31° or 0.54(1) radians.	M1 M1 M1 A1 [4]	Uses $x_1x_2 + y_1y_2 + z_1z_2$ Method for modulus Links everything correctly. co
(ii) **OA** = (3**i** − 4**k)** × (15 ÷ 5) → 9**i** − 12**k** **OB** = (**2i + 3j − 6k**) × (14 ÷ 7) → 4**i** + 6**j** − 12**k**.	M1 A1 A1 [3]	M mark for ×(15 ÷ 5) or ×(14 ÷ 7) A1 for **OA** A1 for **OB**
(iii) **AB** = **b** − **a** = −5**i** + 6**j** → Magnitude of $\sqrt{61}$ or 7.81 → Unit vector of (−5**i** + 6**j**) ÷ $\sqrt{61}$	M1 M1 A1 [3]	Correct use for either **AB** or **BA** Complete method for unit vector. co
9 $y = -x^2 + 8x - 10$		
(i) $\frac{dy}{dx} = -2x + 8$ = 0 when $x = 4$, A is (4, 6) Equation of AB is $y - 6 = 2(x - 4)$ Sim eqns with eqn of curve → $x^2 - 6x + 8 = 0$ or $y^2 - 8y + 12 = 0$ → B (2, 2)	B1 M1A1 M1 M1 A1 A1 [7]	co Sets to 0 and attempt to solve for x. co. Correct form of equation. Eliminates x or y completely Method for quadratic eqn = 0. co (Must not be guessed from diagram)
(ii) $\int -x^2 + 8x - 10\,dx = -\frac{x^3}{3} + 4x^2 - 10x$ Uses his x limits 2 to 4 → 9⅓	B2,1 M1 A1 [4]	3 terms, loses 1 for each error Uses x limits correctly – allow ± co – allow ± (2 must have been correctly found, not guessed)

10 f : $x \mapsto 2x+5$ g : $x \mapsto \dfrac{8}{x-3}$		
(i) $f^{-1} = ½(x-5)$	B1	co
$g^{-1} = \dfrac{8}{x} + 3,\ x \neq 0$	M1 A1 B1√	Attempt at x the subject. co but (f(x) Allow if a linear denominator.
	[4]	
(ii) 	B1 B1 B1	+ve gradient, +ve y intercept +ve gradient, +ve y intercept States, or shows the line $y = x$ as a line of symmetry.
	[3]	
(iii) $fg(x) = \dfrac{16}{x-3} + 5$	B1	co
Forms eqn $\dfrac{16}{x-3} + 5 = 5 - kx$	M1	Must lead to a quadratic
$kx^2 - 3kx + 16 = 0$		
Uses $b^2 = 4ac \rightarrow k = \dfrac{64}{9}$ or 0	M1 A1	Use of $b^2 - 4ac$ even if <0, >0. co Condone ≤
Set of values $0 < k < \dfrac{64}{9}$	A1	co
	[5]	